

Thermodynamic State Vector Interface Contracts and Monadic Exergy Accounting in Planetary Biogeochemical Simulation

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<https://github.com/pascalranoroarijaona/WebOfLife>

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Abstract

Planetary metabolism and macroscopic ecosystems operate under rigorous thermodynamic constraints governed by the First and Second Laws of Thermodynamics. In this preprint, we report the completion of Sprint 022, which formalizes strict TypeScript interface contracts (`src/thermodynamics/types.ts`) and executable monad process flows (`src/thermodynamics/methods.ts`) for internal entropy generation ($\dot{S}_{\text{gen}}$), boundary flux array structures, and Gouy-Stodola exergy destruction accounting ($\dot{I} = T_0 \dot{S}_{\text{gen}}$). By embedding these physical laws directly into compile-time type structures and monad state transition pipelines, the *Web of Life* simulation engine guarantees mathematical closure of mass-energy balances and enforces irreversible thermodynamic dissipation across all simulated biogeochemical cycles (Carbon, Nitrogen, Phosphorus, and Water).

Keywords: Systems Ecology, Thermodynamics, Exergy Destruction, Entropy Generation, Monadic Simulation, Biogeochemical Cycles.

1 Introduction & Theoretical Framework

The *Web of Life* computational architecture models planetary ecosystems as open thermodynamic systems exchanging heat, work, and matter across planetary boundaries. To prevent unphysical behaviors—such as perpetual motion or spontaneous entropy reduction—simulations must enforce strict thermodynamic conservation and dissipation laws at every discrete time step Δt .

1.1 First Law: Conservation of Energy and Mass

The change in internal energy (ΔU) within any subsystem pod or global Earth instance is constrained by net heat transfer (Q), work (W), and boundary mass fluxes carrying specific enthalpies ($\bar{h}_i$):

$$\Delta U = U_{t+\Delta t} - U_t = Q - W + \sum_i \int (\bar{h}_i \dot{m}_i) dt \quad (1)$$

1.2 Second Law: Entropy Generation and Exergy Destruction

All natural biogeochemical transformations are irreversible. The Clausius inequality dictates that the internal entropy generation rate ($\dot{S}_{\text{gen}}$) must remain strictly non-negative:

$$\dot{S}_{\text{gen}} = \frac{dS}{dt} - \sum \left(\frac{\dot{Q}_k}{T_{b,k}} \right) - \sum_{in} \dot{m}_{in} s_{in} + \sum_{out} \dot{m}_{out} s_{out} \geq 0 \quad (2)$$

Invoking the Gouy-Stodola theorem, the exergy destruction rate ($\dot{I}$)—representing the lost work potential due to thermodynamic irreversibilities—is coupled directly to ambient reference temperature T_0 :

$$\dot{I} = T_0 \cdot \dot{S}_{\text{gen}} \geq 0 \quad (3)$$

2 Architecture & Type-Level Contracts

Sprint 022 introduces strict compile-time interface contracts in `src/thermodynamics/types.ts` to govern state vector transformations:

- **IThermodynamicStateVector**: Captures extensive and intensive properties including internal energy U , entropy S , temperature T , pressure P , volume V , and elemental stock masses.
- **IBoundaryFluxVector**: Tracks heat transfer rates $\dot{Q}$, boundary temperatures T_b , mass flow rates, and entropy transfer rates.
- **IEntropyGenerationMetrics**: Encapsulates $\dot{S}_{\text{gen}}$, reference temperature T_0 , exergy destruction rate $\dot{I}$, and a Boolean validation flag for Second Law compliance.

3 Conclusion & Future Directions

Sprint 022 establishes a rigorous thermodynamic foundation for the *Web of Life* engine. By formalizing state vector interfaces and exergy destruction accounting, the framework ensures that macroscopic ecological dynamics emerge from valid microscopic and mesoscopic thermodynamic principles. Future sprints will extend these contracts to multi-pod spatial transport and localized ecological exergy optimization.

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